XChannel

Local radio in six cities: the schedule dashboard and its system design

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Client code and data are private. The demo rebuilds the radio part of the dashboard on the real data model, with made-up data.

What the product does

XChannel runs local radio stations in several cities. Its app plays each station's stream and shows what is on air, who is hosting, and this week's schedule.

My part: the admin dashboard

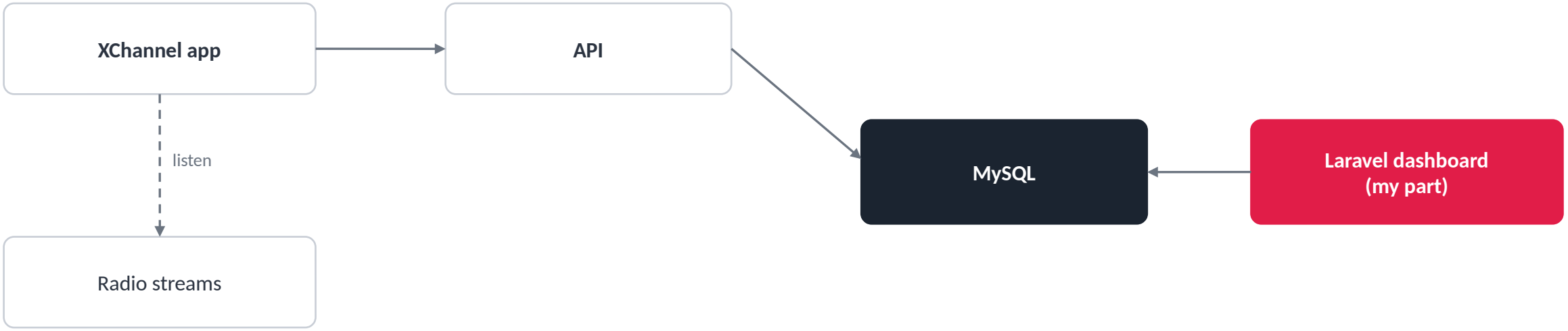
- Stations: city, frequency, stream URL, stream status
- Shows and hosts (master data)
- Each station's weekly schedule, which the app reads
- Podcasts, music chart, video, news: same dashboard, left out of this demo

6 cities, one station each

7 → 1 schedule tables in production, one in the demo

24/7 empty hours play automated music

System overview



One database, two readers

The dashboard writes stations, shows and the schedule. The app reads the same tables through its API.

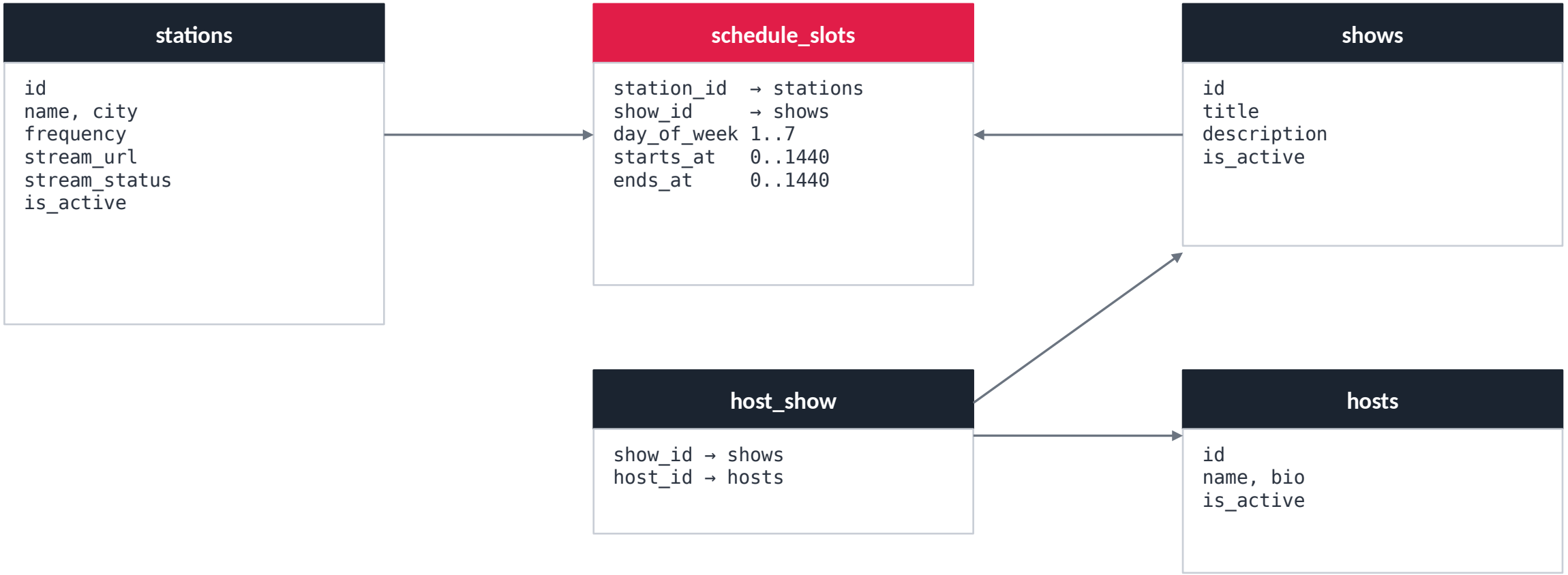
The schedule is the contract

If the schedule overlaps or its format drifts, the app shows the wrong show. The rules belong where the data is saved.

Local time: WIB

Every station is on WIB. The app asks one question: what is on air now?

Data model



Index on (station_id, day_of_week, starts_at) serves both the overlap check and "what is on now".
Deletes are restricted: a show still on the schedule cannot be deleted, only made inactive.

From a table per city to one table

Production

```
radio
  nama_radio, url_radio
  api_jadwal = 'jadwal_radio_woman'

jdwl_radio_bandung
jdwl_radio_surabaya
jdwl_radio_suramadu  ... 7 tables
  judul_siaran, hari_siaran,
  waktu_siaran, penyiar (text)
```

A new station means a new table. The table name is stored as text, and one of them does not match the real table. Solo has a schedule table but no radio row.

Demo

```
stations
  id, name, city, frequency

schedule_slots
  station_id  -> stations
  show_id     -> shows
  day_of_week, starts_at, ends_at

host_show (show_id, host_id)
```

A new station is one row. Foreign keys keep it consistent; one query covers every city.

Gradual migration: fill the new table from the old ones, keep a view per city for the old API, move readers one by one, then drop the old tables.

Time as numbers, not text

What production's day and time columns hold:

'senin-jumat'	'Senin- Jumat'	'Monday-Friday'
'6:00 AM - 10:00 AM'	'08.00-09.00'	'20:00-24:00'

It cannot be sorted, filtered or checked for overlap without parsing free text.

In the demo:

- day_of_week: 1 = Monday … 7 = Sunday (ISO)
- starts_at, ends_at: minutes since midnight
- 07:30 = 450, 24:00 = 1440
- Monday-Friday = five rows, one per day
- The form takes HH:MM only; '6:00 AM', '08.00', '25:00' are refused

Two slots overlap when each starts before the other ends

```
starts_at < :end  AND  ends_at > :start
```

Plain integer comparisons, helped by the index. Touching (10:00 then 10:00) is not an overlap, so shows can run back to back.

The overlap rule on save



A clear message

The error names the day, the time and the show in the way, so the admin knows what to move.

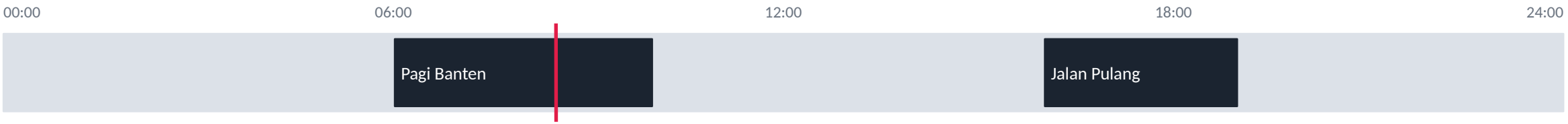
All days or none

One form can schedule Monday to Friday. If one day clashes, nothing is saved (a transaction).

Past midnight

A slot ends by 24:00 at the latest. A Friday 22:00-02:00 show is two slots: Friday until 24:00, Saturday from 00:00.

On air, up next, and empty hours



XChannel Banten, Monday 08:30 (red line = now). Grey parts have no show: the station plays automated music.

On air

Today's slot where start <= current minute < end. One indexed query.

Up next

The first slot after now; if today is done, it looks at the next days, up to a week ahead (Sunday wraps to Monday).

Empty hours

Gaps are worked out from the schedule, not stored. The dashboard shows how many hours a week are still automated music.

The dashboard has a day and time picker, so in a demo we can look at any moment of the week.

Decisions worth talking about

Minutes, not a TIME column

24:00 is not a valid TIME, and integer arithmetic works the same on MySQL, PostgreSQL and SQLite.

Split at midnight

A slot always sits in one day. Overlap and on-air checks stay simple; the cost is two rows for a late-night show.

Airtime per host

Co-hosts each count the full slot, because both work for the whole slot.

Deactivate, not delete

A show still on the schedule, or a host still on a show, cannot be deleted, only made inactive.

Tech stack and why

1 **Laravel 10**

Same as production. Forms, validation, CRUD and migrations are quick; the business rules stay in one service.

2 **MySQL**

The production database. Nothing in the code is tied to it: tests run on SQLite, the online demo on PostgreSQL (Neon, free).

3 **Blade + Tailwind**

The schedule timelines are plain HTML and CSS. An admin dashboard like this needs no JavaScript.

4 **PHPUnit**

Feature tests for every schedule rule: overlap, midnight, on air, up next, gaps.

Keeping the code easy to read

```
public function clash(int $stationId, Day $day,
    int $start, int $end, ?int $ignoreId = null): ?ScheduleSlot
{
    // Two ranges overlap when each one starts before
    // the other ends. Touching (10:00 - 10:00) is fine.
    return ScheduleSlot::with('show')
        ->where('station_id', $stationId)
        ->where('day_of_week', $day->value)
        ->where('starts_at', '<', $end)
        ->where('ends_at', '>', $start)
        ->when($ignoreId, fn ($q) => $q->whereKeyNot($ignoreId))
        ->orderBy('starts_at')
        ->first();
}
```

Rules in one service

ScheduleService: clash, schedule, move, on air, up next, gaps, airtime.

Enums, not numbers

Day (ISO 1-7) and StreamStatus (live, idle, offline).

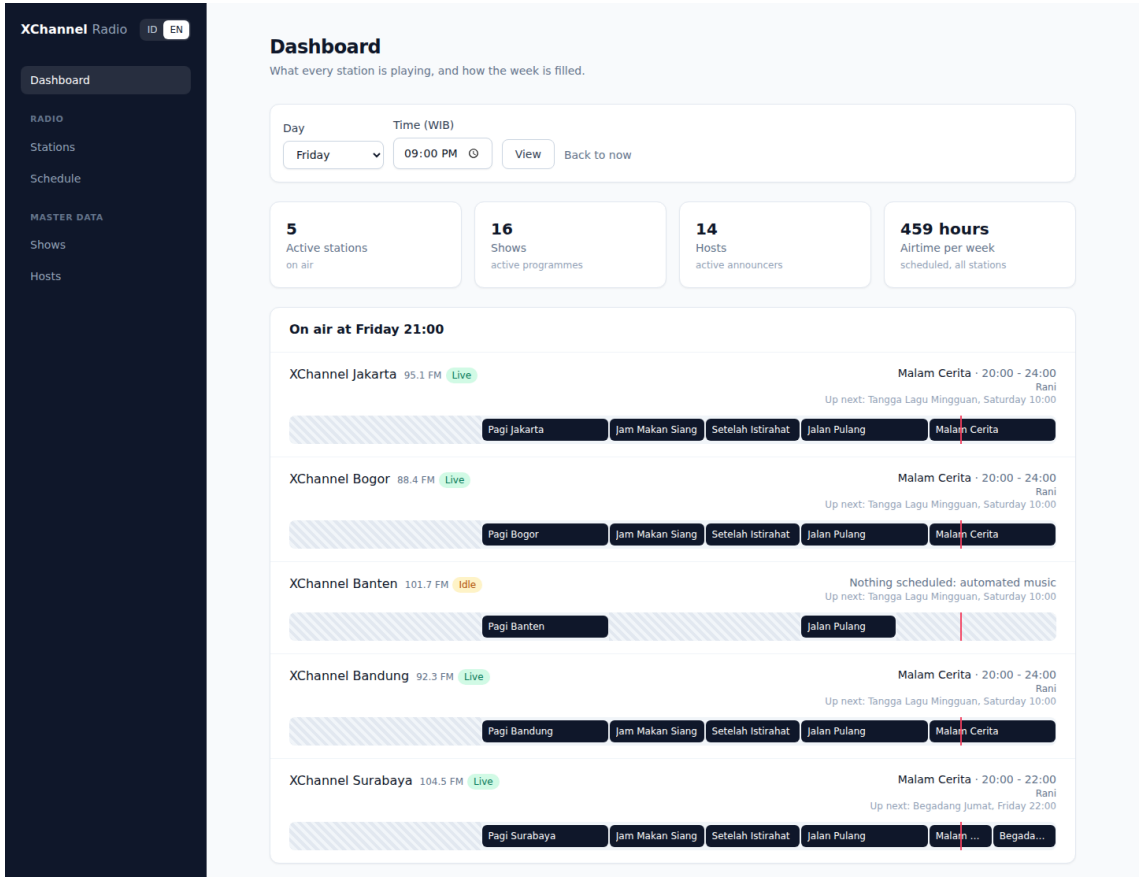
Thin controllers

Validation in FormRequests, time parsing in Clock, rules in the service.

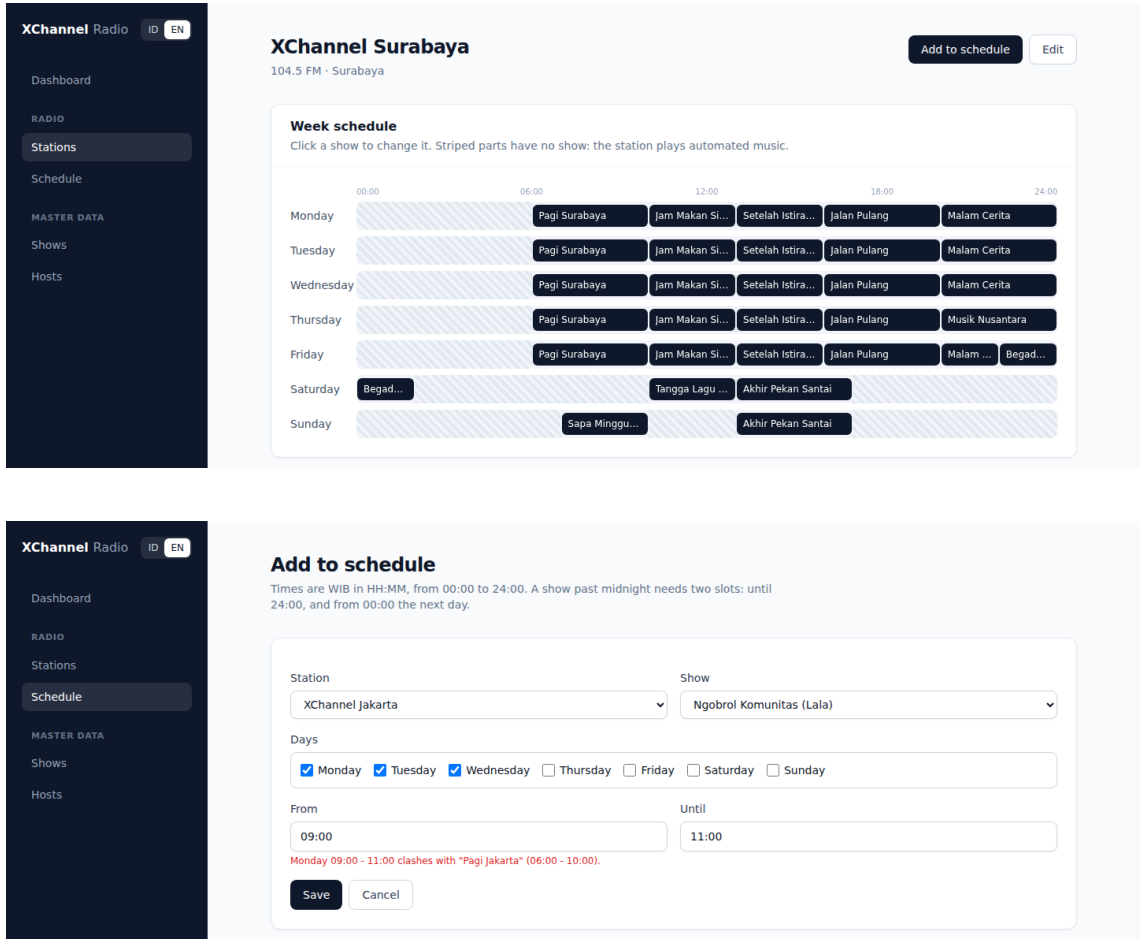
Tests for the rules

19 feature tests, including language and the demo data reset.

The demo app



Dashboard: on air per station and airtime



Top: one station's week. Bottom: an overlapping slot refused.

What I would improve

Move to one table

Fill schedule_slots from the per-city tables, keep views for the old API, move readers over step by step with no downtime.

A time zone per station

If a station opens in WITA or WIT, store its zone on stations and work out "now" per station.

One-off changes

Date-specific overrides (holidays, special broadcasts) on top of the weekly pattern, without editing it.

Stream status from the stream

Set live/idle/offline from a stream health check instead of by hand.

Try it yourself

Live demo	https://xchannel-dashboard-demo.vercel.app
Slides (PDF)	https://xchannel-dashboard-demo.vercel.app/xchannel-walkthrough.pdf
Slides (PPTX)	https://xchannel-dashboard-demo.vercel.app/xchannel-walkthrough.pptx
Demo code	Private GitHub repo, access given during the interview

The dashboard is available in Bahasa Indonesia and English. Demo data can be edited and is restored every 3 hours.